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Electrical connector assembly with coil spring terminal

Abstract

An electrical connector assembly includes an electrical receptacle terminal configured to receive a corresponding electrical plug terminal therein. The electrical receptacle terminal has a helical coil spring. The electrical connector assembly further includes a cable terminal attached to an end of a conductor of an electrical cable. The coil spring is in compressive contact with the cable terminal.

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Background/Summary

TECHNICAL FIELD

(1) This application is directed to an electrical connector assembly that includes a coil spring electrical terminal.

BACKGROUND

(2) Electrical receptacle terminals are typically manufactured from sheet metal using a stamping process and folding process. These receptacle terminals can stub against a mating electrical plug terminal when they are connected if the receptable and plug terminals are not carefully aligned, especially with smaller terminals used to terminate inner conductors of coaxial cables. Therefore, tools to properly orient the receptacle and plug terminals may be required for such terminals. Typically, the receptacle and plug terminals require some overtravel to make the connection, thus requiring more packaging space for a connector assembly using these terminals. Electrical terminals incorporating coil springs usually are not known to use the spring as a contact in an axial application.

BRIEF SUMMARY

(3) According to one or more aspects of the present disclosure, an electrical connector assembly includes an electrical receptacle terminal configured to receive a corresponding electrical plug terminal therein. The electrical receptacle terminal has a helical coil spring. The electrical connector assembly also includes a cable terminal attached to an end of a conductor of an electrical

cable. The coil spring is in compressive contact with the cable terminal.
(4) In one or more embodiments of the electrical connector assembly according to the previous paragraph, the cable terminal defines a spherical sector shaped electrical contact in compressive mechanical contact with the coil spring.
(5) According to one or more aspects of the present disclosure, an electrical connector assembly includes an electrical receptacle terminal having a helical coil spring configured to receive a corresponding electrical plug terminal therein and an insulative housing having a cavity in which the receptacle terminal is disposed.
(6) In one or more embodiments of the electrical connector assembly according to the previous paragraph, a first end of the coil spring has a first portion of the coil spring proximate the first end that is configured to receive the electrical plug terminal and provide a plurality of electrical contact points between the coil spring and the electrical plug terminal.
(7) In one or more embodiments of the electrical connector assembly according to any one of the previous paragraphs, a diameter of coils of the coil spring in the first portion of the coil spring is less than a diameter of coils in the coils spring at the first end.
(8) In one or more embodiments of the electrical connector assembly according to any one of the previous paragraphs, a second end of the coil spring opposite the first end is in compressive mechanical contact with a cable terminal attached to an end of a conductor of an electrical cable.
(9) In one or more embodiments of the electrical connector assembly according to any one of the previous paragraphs, the cable terminal defines a spherical sector shaped electrical contact in compressive mechanical contact with the second end of the electrical receptacle terminal.
(10) In one or more embodiments of the electrical connector assembly according to any one of the previous paragraphs, a longitudinal axis of the electrical cable is arranged generally perpendicularly to a longitudinal axis of the coil spring.
(11) In one or more embodiments of the electrical connector assembly according to any one of the previous paragraphs, coils of the coil spring in a second portion of the coil spring outside of the electrically conductive sleeve are touching one another when the coil spring is in a relaxed condition.
(12) In one or more embodiments of the electrical connector assembly according to any one of the previous paragraphs, the electrically conductive sleeve defines a tab feature configured to retain the electrically conductive sleeve and the coil spring within the cavity.
(13) In one or more embodiments of the electrical connector assembly according to any one of the previous paragraphs, the coil spring is a first coil spring. The assembly further includes a second coil spring that is partially disposed within the electrically conductive sleeve and in compressive mechanical contact with a cable terminal attached to an end of a conductor of an electrical cable.
(14) In one or more embodiments of the electrical connector assembly according to any one of the previous paragraphs, coils of the second coil spring outside of the electrically conductive sleeve are touching one another when the coil spring is in a relaxed condition.
(15) According to one or more aspects of the present disclosure, an electrical connector assembly includes an electrical receptacle terminal configured to receive a corresponding electrical plug terminal therein, a cable terminal attached to an end of a conductor of an electrical cable, a coil spring having a first end attached to the electrical receptacle terminal a second end in compressive mechanical contact with the cable terminal, and an insulative housing having a cavity in which the receptacle terminal and coil spring are disposed.
(16) In one or more embodiments of the electrical connector assembly according to the previous paragraph, coils of the coil spring located proximate the second end are touching one another when the coil spring is in a relaxed condition.
(17) In one or more embodiments of the electrical connector assembly according to any one of the previous paragraphs, a portion of the receptacle terminal is disposed within a first portion of the coil spring. Coils of the coil spring in a second portion of the coil spring in which the electrical

receptacle terminal is not disposed are touching one another when the coil spring is in a relaxed condition.
(18) In one or more embodiments of the electrical connector assembly according to any one of the previous paragraphs, the portion of the electrical receptacle terminal received within the coil spring is rod-shaped.
(19) In one or more embodiments of the electrical connector assembly according to any one of the previous paragraphs, the cable terminal defines a spherical sector shaped electrical contact.
(20) In one or more embodiments of the electrical connector assembly according to any one of the previous paragraphs, the cable terminal is an electrically conductive sphere attached to the end of a conductor of an electrical cable.
(21) In one or more embodiments of the electrical connector assembly according to any one of the previous paragraphs, the first end of the coil spring is in compressive contact with rod-shaped portion of the electrical receptacle terminal.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The present invention will now be described by way of example with reference to the accompanying drawings, in which:
(2) FIG. **1** shows a cross-section view of an electrical connector assembly with an electrical receptacle terminal having a helical coil spring configured to receive a corresponding electrical plug terminal according to some embodiments;
(3) FIG. **2** shows a close up cross-section view of the interface between the helical coil spring of the electrical receptacle terminal of FIG. **1** and the corresponding electrical plug terminal according to some embodiments;
(4) FIG. **3** shows a perspective view of the helical coil spring of the electrical receptacle terminal of FIG. **1** within an electrically conductive sleeve according to some embodiments;
(5) FIGS. **4**A and **4**B show various steps in a process of forming the electrically conductive sleeve of FIG. **3** around the helical coil spring according to some embodiments;
(6) FIG. **5** shows a cross-section view of the electrical connector assembly of FIG. **1** further including an electrically conductive sphere terminating a conductor of an electrical cable according to some embodiments;
(7) FIG. **6** shows a cross-section view of the electrical connector assembly of FIG. **1** interfacing with a corresponding electrical connector assembly including the corresponding electrical plug terminal according to some embodiments;
(8) FIG. **7** shows a close up cross-section view of the interface between the helical coil spring of the electrical receptacle terminal of FIG. **1** and the electrically conductive sphere according to some embodiments;
(9) FIG. **8** shows a cross-section view of an electrical connector assembly with an electrical receptacle terminal having a first helical coil spring configured to receive a corresponding electrical plug terminal and a second helical coil spring configured to mechanically and electrically contact a spherical sector shaped electrical contact according to some embodiments.
(10) FIG. **9** shows a cross-section view of an electrical connector assembly with an electrical receptacle terminal having a helical coil spring configured to mechanically and electrically contact a spherical sector shaped electrical contact according to some embodiments;
(11) FIG. **10**A shows a perspective view of the electrical receptacle terminal of FIG. **9** according to some embodiments;
(12) FIG. **10**B shows a cross-section view of the electrical receptacle terminal of FIG. **9** according to some embodiments;

(13) FIG. **11** shows a cross-section view of the electrical connector assembly of FIG. interfacing with a corresponding electrical connector assembly according to some embodiments; and
(14) FIG. **12** shows a perspective view of an electrical receptacle terminal according to some embodiments.

DETAILED DESCRIPTION

(15) FIGS. **1-7** show a first non-limiting example of a right-angled electrical connector assembly, hereafter referred to as the assembly **100**. The assembly **100** includes an electrical receptacle terminal **102** in the form of a helical coil spring **104** that is configured to receive a corresponding electrical plug terminal **106**, shown here in the form of a cylindrical pin, of a mating electrical connector assembly **108**. The electrical receptacle terminal **102** is disposed within a cavity **110** of an insulative housing **112** which is surrounded by an electrical shield terminal **114**. The electrical receptacle terminal **102** is configured to terminate a central conductor **116** of an electrical cable **118** and the electrical shield terminal **114** terminates a shield conductor **120** of the electrical cable **118**. In the present embodiment, the electrical cable **118** is a coaxial cable.
(16) As best shown in FIG. **2**, a first end **122** of the coil spring **104** has an opening **124** with a first diameter **126**. A first portion **128** of the coil spring **104** near the first end **122** has a second diameter **130** that is reduced from, i.e., smaller than, the first diameter **126**. This first portion **128** is configured to provide compressive contact between the receptacle terminal **102** and the corresponding plug terminal **106**. The coil spring **104** not only provides a compressive contact force between the receptacle and plug terminals **102**, **106**, the turns of the coil spring **104** also provide a plurality of redundant contact points between the receptacle and plug terminals **102**, **106**, thereby improving the robustness of the electrical connection between the receptacle and plug terminals **102**, **106**.
(17) As shown in FIG. **3**, the first portion **128** of the coil spring **104** is contained within a conductive sleeve **132** that is at least partially radially wrapped around the first portion **128**. FIGS. **4**A and **4**B illustrate a process of forming conductive sleeve **132** from a terminal preform **134** stamped from flat sheet metal as shown in FIG. **4**A and then rolled around the coil spring **104** to form the conductive sleeve **132** in FIG. **4**B. This conductive sleeve **132** improves the electromagnetic performance of the receptacle terminal **102** by covering the gaps **136** in the coil spring in and around the first portion **128**.
(18) A second end **138** of the coil spring **104** arranged opposite the first end **122** is in compressive mechanical and electrical contact with an electrical contact **140** connected to an end of the central conductor **116** of the coaxial cable **118**. The electrical contact **140** may define a spherical sector shaped electrical contact. In alternative embodiments, the electrical contact **140** may have a flat or tubular shape. In this particular example, the spherical sector shape of the electrical contact **140** is provided by an electrically conductive ball or sphere **142** that is attached, e.g., welded, to the end of the central conductor **116**. The second end **138** has a third diameter **144** that is larger than the first diameter **126** but less than a diameter **146** of the sphere **142**. As shown in FIG. **5**, the second end **138** presses against the sphere **142** in the longitudinal or X direction, the coil spring **104** is thereby compressed and exerts a compressive contact force between the second end **138** of the coil spring **104** and the sphere **142**, thereby providing a robust electrical connection between the receptacle terminal **102** and the central conductor **116**. As shown in FIG. **7**, the coil spring **104** may flex in the lateral or Y direction to accommodate some longitudinal misalignment between the coil spring **104** and the sphere **142**.
(19) Returning now to FIG. **3**, the coils of the coil spring **104** in a second portion **148** of the coil spring **104** located proximate the second end **138** and extending from the conductive sleeve **132** touch one another when the coil spring **104** is in a relaxed condition. This feature wherein the coils touch one another has also been found to improve the electromagnetic performance of the receptacle terminal **102**.
(20) Although the first and second ends **122**, **138** of the coil spring **104** are illustrated as having a

blunt cut, other embodiments of the coil spring **104** may be envisioned in which the first end **122**, the second end **138**, or both are flush cut.

(21) FIG. **8** shows another non-limiting example of a right-angled electrical connector assembly, hereafter referred to as the assembly **200**. In this example, which is similar to assembly **100**, the coil spring is divided into two separate coil springs **204**A, **204**B that are electrically and mechanically connected by a conductive sleeve **232**. The first coil spring **204**A has a first end **222** that is configured to receive the plug terminal **106** of the mating electrical connector assembly **108** and the second coil spring **204**B is configured to contact the electrical contact **140** connected to the end of the central conductor **116** of the coaxial cable **118**. The coils of the second coil spring **204**B in a portion **248** of the second coil spring **204**B located proximate the second end **238** and extending from the conductive sleeve **232** touch one another when the second coil spring **204**B is in a relaxed condition. The second end **238** is in compressive contact with the sphere **142** of the electrical contact **140**. The diameter **244** of the second end **238** is less than a diameter **146** of the sphere **142**. The second coil spring **204**B may flex in the lateral or Y direction to accommodate some longitudinal misalignment between the second coil spring **204**B and the sphere **142**.

(22) FIGS. **9-11** show yet another non-limiting example of a right-angled electrical connector assembly, hereafter referred to as the assembly **300**. In this example, which is similar to assemblies **100** and **200**, the receptacle terminal **302** shown in FIGS. **10**A and **10**B has a conventional stamped and rolled tubular portion **350** that is configured to receive the plug terminal **106**. The receptacle terminal **302** includes a coil spring **304** and a rod-like portion **352** extending from the tubular portion **350** is received in a first end **322** of the coil spring **304**. The coil spring **304** extends between the first end **322** and a second end **338** opposite the first end **322**. The coils of the coil spring **304** in a portion **348** of the coil spring **304** located proximate the second end **338** and extending past the rod-like portion **352** touch one another when the coil spring **304** is in a relaxed condition.

(23) In the example illustrated in FIG. **9**, the second end **338** of the coil spring **304** is in compressive contact with an electrical terminal **340** that is connected, e.g., crimped, to the end of the central conductor **116** of a coaxial cable **118**. The electrical terminal **340** defines a spherical sector shaped surface **342** that is in compressive contact with the second end **338** of the coil spring **304**. The portion **348** of the coil spring **304** has a diameter **344** that is larger than a diameter **326** of the rest of the coil spring **304** but less than a diameter **146** of the spherical sector shaped surface **342**. In an alternative example illustrated in in FIG. **11**, the second end **338** of the coil spring **304** is in compressive contact with the sphere **142** of the electrical contact **140** that is connected, e.g., welded, to the end of the central conductor **116** of a coaxial cable **118**.

(24) FIG. **12** shows an alternative coil spring **304′** in which the portion **348** of the coil spring **304′** has a diameter **344′** that is generally the same as a diameter **326** of the rest of the coil spring **304′**.

(25) The coil spring **304** or **304′** may flex in the lateral or Y direction to accommodate some longitudinal misalignment between the coil spring **304** or **304′** and the sphere **142** or spherical sector shaped surface **342**.

(26) Other embodiments may be envisioned in which the conductive sphere **142** welded to the central conductor **116** and the electrical terminal **340** that is crimped to the central conductor **116** are interchanged.

(27) While the illustrated examples are right angled electrical connectors, alternative embodiments may be envisioned for straight electrical connectors or other non-perpendicular arrangements between the plug terminal and the electrical cable. In addition, other alternative embodiments of electrical connector assemblies employed in other receptacle terminal-plug terminal configurations may also be envisioned.

(28) While the invention has been described with reference to an exemplary embodiment(s), it will be understood by those skilled in the art that various changes may be made, and equivalents may be substituted for elements thereof without departing from the scope of the invention. In addition,

many modifications may be made to adapt a particular situation or material to the teachings of the invention without departing from the essential scope thereof. Therefore, it is intended that the invention is not limited to the disclosed embodiment(s), but that the invention will include all embodiments falling within the scope of the appended claims.

(29) As used herein, 'one or more' includes a function being performed by one element, a function being performed by more than one element, e.g., in a distributed fashion, several functions being performed by one element, several functions being performed by several elements, or any combination of the above.

(30) It will also be understood that, although the terms first, second, etc. are, in some instances, used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first contact could be termed a second contact, and, similarly, a second contact could be termed a first contact, without departing from the scope of the various described embodiments. The first contact and the second contact are both contacts, but they are not the same contact.

(31) The terminology used in the description of the various described embodiments herein is for the purpose of describing particular embodiments only and is not intended to be limiting. As used in the description of the various described embodiments and the appended claims, the singular forms "a", "an" and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term "and/or" as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It will be further understood that the terms "includes," "including," "comprises," and/or "comprising," when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

(32) As used herein, the term "if" is, optionally, construed to mean "when" or "upon" or "in response to determining" or "in response to detecting," depending on the context. Similarly, the phrase "if it is determined" or "if [a stated condition or event] is detected" is, optionally, construed to mean "upon determining" or "in response to determining" or "upon detecting [the stated condition or event]" or "in response to detecting [the stated condition or event]," depending on the context.

(33) Additionally, while terms of ordinance or orientation may be used herein these elements should not be limited by these terms. All terms of ordinance or orientation, unless stated otherwise, are used for purposes distinguishing one element from another, and do not denote any particular order, order of operations, direction or orientation unless stated otherwise.

Claims

1. An electrical connector assembly, comprising: an electrical receptacle terminal having a helical coil spring configured to receive a corresponding electrical plug terminal therein; and an insulative housing having a cavity in which the receptacle terminal is disposed, an end of the coil spring being in compressive mechanical contact with a cable terminal attached to an end of a conductor of an electrical cable, said cable terminal defining a spherical sector shaped electrical contact in compressive mechanical contact with the electrical receptacle terminal.
2. The electrical connector assembly according to claim 1, wherein another end of the coil spring has a first portion of the coil spring that is configured to receive the electrical plug terminal and provide a plurality of electrical contact points between the coil spring and the electrical plug terminal.
3. The electrical connector assembly according to claim 1, wherein a diameter of coils of the coil spring in the first portion of the coil spring is less than a diameter of coils in the coil spring at the another end.

4. The electrical connector assembly according to claim 1, wherein a longitudinal axis of the electrical cable is arranged generally perpendicularly to a longitudinal axis of the coil spring.
5. The electrical connector assembly according to claim 1, wherein a first portion of the coil spring is contained within an electrically conductive sleeve at least partially radially surrounding the coil spring.
6. The electrical connector assembly according to claim 5, wherein the electrically conductive sleeve defines a tab feature configured to retain the electrically conductive sleeve and the coil spring within the cavity.
7. The electrical connector assembly according to claim 5, wherein the coil spring is a first coil spring, wherein the electrical connector assembly further comprises a second coil spring partially disposed within the electrically conductive sleeve and in compressive mechanical contact with a cable terminal attached to an end of a conductor of an electrical cable.
8. The electrical connector assembly according to claim 7, wherein coils of the second coil spring outside of the electrically conductive sleeve are touching one another when the coil spring is in a relaxed condition.
9. An electrical connector assembly comprising: an electrical receptacle terminal having a helical coil spring configured to receive a corresponding electrical plug terminal therein, a first portion of the coil spring being contained within an electrically conductive sleeve at least partially radially surrounding the coil spring and coils of the coil spring in a second portion of the coil spring outside of the electrically conductive sleeve are touching one another when the coil spring is in a relaxed condition; and an insulative housing having a cavity in which the receptacle terminal is disposed.
10. An electrical connector assembly, comprising: an electrical receptacle terminal configured to receive a corresponding electrical plug terminal therein; a cable terminal attached to an end of a conductor of an electrical cable; a coil spring having a first end attached to the electrical receptacle terminal and a second end in compressive mechanical contact with a spherical sector shaped electrical contact of the cable terminal; and an insulative housing having a cavity in which the receptacle terminal and coil spring are disposed.
11. The electrical connector assembly according to claim 10, wherein coils of the coil spring located proximate the second end are touching one another when the coil spring is in a relaxed condition.
12. The electrical connector assembly according to claim 10, wherein a portion of the receptacle terminal is disposed within a first portion of the coil spring and wherein coils of the coil spring in a second portion of the coil spring in which the electrical receptacle terminal is not disposed are touching one another when the coil spring is in a relaxed condition.
13. The electrical connector assembly according to claim 12, wherein the portion of the electrical receptacle terminal received within the coil spring is rod-shaped.
14. The electrical connector assembly according to claim 10, wherein the cable terminal is an electrically conductive sphere attached to the end of a conductor of an electrical cable.
15. The electrical connector assembly according to claim 10, wherein the first end of the coil spring is in compressive contact with a rod-shaped portion of the electrical receptacle terminal.